

Task 1 – ADR Identification Basics

What I Learned About Adverse Drug Reactions and Pharmacovigilance

Introduction to Pharmacovigilance

From the tutorial, I understood that pharmacovigilance is mainly concerned with the safety of medicines. It involves identifying, monitoring, assessing and preventing unwanted effects that may occur when medicines are used.

Pharmacovigilance is important because a medicine may show certain unwanted effects only after it is used by a large number of people. Reporting these reactions helps healthcare professionals and regulatory authorities understand the safety profile of medicines and take appropriate action when required.

Classification of ADRs

I learned that adverse drug reactions can occur due to different reasons and mechanisms. Some of the important types discussed were:

- **Pharmacological ADRs:** These are related to the normal action of a drug, particularly when the drug effect becomes excessive.
 - **Pathological ADRs:** These are associated with pathological or disease-related changes that may occur during drug treatment.
 - **Genotoxic ADRs:** These involve damage to genetic material and may result in mutations or other long-term effects.
 - **Allergic ADRs:** These occur when the body's immune system reacts to a particular medicine.
 - **Idiosyncratic ADRs:** These are unusual and unpredictable reactions that cannot be easily explained by the normal action of the drug.
 - **Pharmacogenetic ADRs:** These occur because genetic differences can make some people respond differently to a particular medicine.
 - **Teratogenic ADRs:** These refer to harmful effects of medicines on a developing fetus, particularly when certain drugs are taken during pregnancy.
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Hypersensitivity Reactions

The tutorial also introduced the four types of hypersensitivity reactions:

- **Type I:** This is an immediate reaction and can include conditions such as urticaria, angioedema and anaphylaxis.
- **Type II:** In this type, antibodies react with cells and may cause cell destruction.
- **Type III:** This occurs because of immune-complex formation, which can result in inflammation.
- **Type IV:** This is a delayed reaction and is mainly mediated by T cells. Some drug-related skin reactions can occur through this mechanism.

Understanding these types helps in recognizing whether an adverse reaction may have an allergic or immune-related mechanism.

Teratogenic Drugs and Pregnancy

I also learned about the traditional A, B, C, D and X categories used to describe the potential risks of medicines during pregnancy:

- **Category A:** Studies have not shown a risk to the fetus.
- **Category B:** No significant risk was demonstrated in animal studies, although human data may be limited.
- **Category C:** A risk cannot be ruled out, so the medicine should only be used when the expected benefit justifies the potential risk.
- **Category D:** There is evidence of fetal risk, but the medicine may still be considered when the benefits are greater than the risks.
- **Category X:** The risk to the fetus is considered to outweigh any possible benefit, so these medicines should not be used during pregnancy.

Note: I also learned that these letter categories are a traditional FDA system and have been replaced by more detailed pregnancy and lactation information in current FDA labeling.

Causality Assessment

One important part of ADR identification is deciding whether the reaction could actually be related to the suspected medicine. This is known as causality assessment. The tutorial introduced three approaches:

- **Bradford Hill Criteria:** These criteria are used to consider whether an observed relationship is likely to be causal. Factors such as timing, consistency, strength of the association and biological plausibility are considered.
 - **WHO-UMC Scale:** This system is used to assess the relationship between a suspected drug and an adverse reaction. The relationship can be classified into categories such as certain, probable/likely, possible, unlikely and other categories when the available information is insufficient.
 - **Naranjo Algorithm:** This is a questionnaire-based method where a series of questions are used to determine how likely it is that a particular medicine caused the reaction.
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How ADRs Are Reported in India

I learned that a suspected ADR should be reported even when we are not completely sure that the medicine caused it. The available patient, medicine and reaction information is recorded in the appropriate CDSCO ADR reporting form.

ADR reporting can also be done through the ADR PvPI mobile application, which makes reporting more convenient. The report generally contains details such as the patient's information, suspected medicine, description of the reaction, relevant medical history and the timeline between starting the medicine and developing the reaction.

Pharmacovigilance Programme of India (PvPI)

The Pharmacovigilance Programme of India (PvPI) is the national programme for monitoring adverse drug reactions in India. ADR reports are collected through Adverse Drug Reaction Monitoring Centres (AMCs) and are assessed and forwarded through the PvPI system. The information collected can help identify possible safety concerns and improve the safe use of medicines.
